

SIRATUL ISLAM

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Education

Shahjalal University of Science and Technology

Bachelor of Science (Hons) in Physics

Aug. 2023 – Aug 2027

Sylhet, Bangladesh

Experience

The Zephyr RTOS Project (Linux Foundation)

Remote

Biometrics, Authentication, and FIDO2 Subsystems Maintainer

Feb. 2026 – Present

- Author and Maintainer of Zephyr's Biometrics driver class (PR #100139)
- Author and Maintainer of Zephyr's FIDO2 & Authentication subsystems (PR #104327)
- Review community contributions on various subsystems
- Technologies: Embedded C, Subsystem Architecture, Device Drivers, Device Tree, USB, PSA Crypto

Contributor (Triage)

Jun. 2025 – Present

- Zephyr Contributor with Triage access and 20+ merged PRs across subsystems, drivers, fixes, and BSPs
- Develop device drivers and board support packages serving embedded developers globally
- Review community contributions and provide technical guidance on driver architecture and device tree bindings

The Linux Kernel

Remote

Driver Maintainer & Reviewer, Industrial I/O (IIO) Subsystem

Mar. 2026 – Present

- **Maintained Drivers:** ST VL53L1X (Time-of-Flight sensor), QST QMC5883L (Magnetometer)
- Authored the drivers and Device Tree bindings and used latest Kernel APIs
- Review patches in linux-iio on lore.kernel.org, and collaborate with contributors

Shahjalal University of Science and Technology

Jun. 2025 – Present

Research Assistant - Department of Electrical and Electronic Engineering

Sylhet, Bangladesh

- Developed a Zephyr based smart notification/timer/clock system for EEE department with OTA and Firmware signing
- Developing IoT-enabled smart control systems and radar-based occupancy detection for government energy optimization
- Building RFID-based attendance management systems targeting rural educational institutions in Bangladesh

Hackules Inc.

Feb. 2024 – Jun. 2025

Software Engineer

Remote

- Led full-stack/frontend development for educational platforms (Opedemy, Teachers Today) serving 1000+ active users

Selected Projects

Wireless Smart Display System | Zephyr, ESP32, OTA, Display, MQTT, MCUBoot

[source code](#)

- Wrote the official Zephyr HUB12 Display driver.
- Implemented OTA, including progress indicator and used MCUBoot with firmware signing for enhanced security
- Implemented multiple switchable clock-faces and advanced scheduling features for notifications and timer.

STM32 BME280 HAL Driver | C, STM32, I2C, Bare-Metal

[source code](#)

- Developed a custom Hardware Abstraction Layer (HAL) driver in C for the Bosch BME280 environmental sensor
- Implemented reliable I2C communication protocols to parse temperature, humidity, and pressure telemetry

Publications

1. (Under Preparation) S. Islam, M. Rasedujjaman, "BASE: A Capability-Driven Biometric Subsystem for Embedded Real-Time Systems," Target - *Journal of System Architecture, Elsevier*.
2. (Under Preparation) S. Islam, M. Rasedujjaman, "FIDO2/CTAP2.1 on Zephyr RTOS: Design and Multi-Platform Evaluation of an Open-Source FIDO2 Authenticator," Target - *IEEE Transactions on Computers*

Technical Skills

Specializations : Zephyr RTOS, Linux Kernel, FIDO2/CTAP2, Embedded Biometrics, IoT Security & Hardware Auth

Embedded Systems: Zephyr RTOS, FreeRTOS, Linux Kernel, STM32 HAL, ESP-IDF, Embedded Linux, ROS2

Platforms & Protocols : STM32, ESP32, nRF52, Raspberry Pi, MQTT, BLE, Wi-Fi, LoRa, USB, NFC, I2C, SPI, UART

Development Tools: Embedded C/C++, Rust, STM32CubeIDE, CMake, PlatformIO, OpenOCD, KiCAD, Git

Full-Stack Web: C#, TypeScript, .NET, SvelteKit, Next.js, PostgreSQL, Docker, Podman